# Zhe Wang

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yoyostudy.github.io

## Research Interest

I am broadly interested in large language models. In particular, I am interested in model architecture (model depth, looped models). In the meantime, I care about LLM security, especially the potential threats that frontier LLMs may pose to society.

## Education

**Simon Fraser University** — Vancouver, Canada
*Ph.D. in Computer Science* — 2025 - Present
- Advisor: Dr. Linyi Li
- Research area: Large Reasoning Models and Trustworthy AI

**Simon Fraser University** — Vancouver, Canada
*M.Sc. in Computer Science* — 2022 - 2024

**The Chinese University of Hong Kong, Shenzhen** — Shenzhen, China
*B.Sc. in Mathematics* — 2018 - 2022

## Publications

1. **Zhe Wang**, Mohamad A. Tayebi. AutoRed: Automated Attack Scenario Generation Framework for Red Teaming of LLMs. *IEEE International Conference on Big Data (BigData), Industry & Governance Track, 2024.*

## Internships

**RBC Borealis** | Vancouver, Canada — 2026.01 – 2026.09
*Mitacs Research Intern*
- Identified a state-supervision mismatch in looped models such as tiny recursive models (TRMs): with deep supervision and gradient detachment, early steps are trained on answers they cannot yet derive.
- Designed adaptive supervision that trains each step only on what the model can already derive, improving Sudoku-Extreme accuracy by 7.9 points with up to 1.79× faster training.
- Resulted in a manuscript submitted to ICLR 2027 for peer review.

**SAP Canada Inc.** | Vancouver, Canada — 2024.09 – 2025.08
*Research and Development Intern*
- Post-trained 8B-parameter language models (SFT, LoRA, DPO, GRPO) for software license-permission extraction and classification.
- Developed a Retrieval-Augmented Generation (RAG) system using LangGraph, fully integrated with SAP AI Core for domain-specific retrieval.
- Improved structured output reliability by transitioning from JSON prompting to function-calling with Pydantic, reducing schema violations and hallucinations.
- Deployed 4-bit and 8-bit quantized models with Ollama on GPU (GCP) and CPU to support lightweight demos.

**Hanglok-Tech** | Shenzhen, China — 2023.07 – 2023.09
*Machine Learning Research Intern*
- Improved robotic control performance in Panda-Gym using DDPG (Stable Baselines3).
- Applied GAIL in Isaac Gym for real-time humanoid motion imitation.

## Service

Conference Reviewer: WWW, EMNLP